ODE Definitions and Terminology

- A **first-order linear IVP** is $\boldsymbol{y}' = \boldsymbol{A}(t)\boldsymbol{y} + \boldsymbol{f}(t)$ with $\boldsymbol{y}(0) = \boldsymbol{y}_0$.
  - The **general solution (GS)** is $\boldsymbol{y}(t) = \boldsymbol{y}_c(t) + \boldsymbol{y}_p(t)$.
  - The **complementary solution** $\boldsymbol{y}_c$ (containing constants $\boldsymbol{c}$) satisfies $\boldsymbol{y}_c' = \boldsymbol{A}(t)\boldsymbol{y}_c + \boldsymbol{0}$.
  - A **particular solution** $\boldsymbol{y}_p$ (no undetermined constants) satisfies $\boldsymbol{y}_p' = \boldsymbol{A}(t)\boldsymbol{y}_p + \boldsymbol{f}(t)$.
  - The constants $\boldsymbol{c}$ in the GS are determined by solving the LS $\boldsymbol{y}_c(0) + \boldsymbol{y}_p(0) = \boldsymbol{y}_0$.
- **Constant Coefficient** means $\boldsymbol{A}$ is independent of t.
  - Eigenpairs satisfying $\boldsymbol{A}\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i$ give solutions $e^{\lambda_i t}\boldsymbol{v}_i$ of $\boldsymbol{y}' = \boldsymbol{A}\boldsymbol{y}$
  - With a complete LI set of eigenvectors $\boldsymbol{y}_c(t) = \sum_{i=1}^{n} c_i e^{\lambda_i t}\boldsymbol{v}_i$.
  - **Undetermined Coefficients** guesses a $\boldsymbol{y}_p$ form by looking at $\boldsymbol{f}(t)$. The coefficients are determined by substituting into the ODE $\boldsymbol{y}_p' = \boldsymbol{A}(t)\boldsymbol{y}_p + \boldsymbol{f}(t)$.

Complex Eigenvalues and Eigenvectors $y' = Ay$

Take the real matrix

$$\boldsymbol{A} = \begin{pmatrix} -1 & -2 \\ 2 & -1 \end{pmatrix} \qquad \det(\boldsymbol{A} - \lambda I) = (\lambda + 1)^2 + 4 = 0$$

Its eigenvalues and eigenvectors are

$$\lambda_1 = -1 + 2i \qquad \boldsymbol{v}_1 = \begin{pmatrix} 1 \\ -i \end{pmatrix} \qquad \lambda_2 = -1 - 2i \qquad \boldsymbol{v}_2 = \begin{pmatrix} 1 \\ i \end{pmatrix}$$

The complex general solution is

$$\boxed{\boldsymbol{y}(t) = c_1 e^{(-1+2i)t} \begin{pmatrix} 1 \\ -i \end{pmatrix} + c_2 e^{(-1-2i)t} \begin{pmatrix} 1 \\ i \end{pmatrix}} \qquad c_1, c_2 \in \mathbb{C}$$